

## Analysis First-Year Exam, January 2023, Part 2

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Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let  $f$  be a real-valued function on  $[0, 1]$  and consider the statements below. Prove that (ii) implies (i); and show by example that (i)  $\Rightarrow$  (ii) can fail.
  - (i) for each positive integer  $n$  the function  $f$  is Riemann-integrable on  $[1/n, 1]$  and the sequence of integrals  $\int_{1/n}^1 f$  converges to a real number as  $n \rightarrow \infty$ ;
  - (ii) the function  $f$  is Riemann-integrable on  $[0, 1]$ .
2. Let  $X$  be a metric space on which  $(f_n)$  and  $(g_n)$  are uniformly convergent sequences of continuous real-valued functions.
  - (i) Prove that if  $X$  is compact then  $(f_n g_n)$  is uniformly convergent.
  - (ii) Show that the conclusion in (i) can fail when  $X$  is not compact.

3. Let  $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$  be continuous. By first considering polynomials in two variables, prove that

$$\int_0^1 \left( \int_0^1 f(x, y) dx \right) dy = \int_0^1 \left( \int_0^1 f(x, y) dy \right) dx.$$

4. Prove that if  $(f_n)$  is a sequence of measurable functions such that  $f_n \rightarrow f$  on some measure space, then  $f$  is measurable. Hence, or otherwise, show that if  $f : \mathbb{R} \rightarrow \mathbb{R}$  is differentiable then its derivative  $f'$  is Borel measurable.
5. In each of the following cases, give (if possible and with justification) a sequence  $(f_n)$  of Lebesgue integrable functions from  $\mathbb{R}$  to  $\mathbb{R}$  such that, where  $\lambda$  denotes Lebesgue measure:
  - (i)  $f_n \rightarrow 0$  pointwise but  $\int f_n d\lambda \rightarrow 1$  as  $n \rightarrow \infty$ ;
  - (ii)  $f_n \rightarrow 0$  pointwise but  $\int f_n d\lambda \rightarrow \infty$  as  $n \rightarrow \infty$ .